

# Twenty questions for Dr. Burghardt

## *What the speaker notes almost say*

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Your speaker notes run to roughly six thousand words across thirty-two slides, and in several places they are better teaching than anything on the slides themselves — the account of why dry, water and oil objectives top out at 0.95, 1.20 and 1.40 is the clearest I have read anywhere, because it is the only one that explains the numbers mechanically rather than optically.

That corpus is a sample of what you know, not the whole of it. These questions are aimed at the parts you did not write down because they are obvious to you after thirty-eight years, and which are exactly the parts a graduate student cannot get from a textbook. Each is anchored to the slide and note it came from, so you can check my reading.

*Most can be answered in a sentence. Answer in the space below each, in any order, and skip anything that is not worth your time — a partial return is far more useful than none. Where a question asks you to rank things, a rough ordering is plenty.*

### SECTION 1 – NUMERICAL APERTURE AND THE OBJECTIVE ITSELF

#### **Q1** slide 22, note

You left “What does TIR stand for???” in the notes. Is that a reminder to yourself, or a question you deliberately put to the room?

*If it is a question you pose to students, we would like to keep it as one rather than answer it for them.*

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#### **Q2** slide 22, note

The sentence “High NA lenses with high mag can resolve more than low mag lenses with high NA” reads ambiguously as written. What is the point you want students to take from it?

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#### **Q3** slide 22, note

Your explanation that the practical limits are mechanical — the front lens cannot touch the coverslip, and on a 100× it is barely a millimetre across — is the best thing in the notes. Do you show students an actual front element to make it land, or is it purely verbal?

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**Q4** slide 22, note

You mention in passing that Olympus made a 100×/1.65. What does it require in practice, and why does essentially nobody use one?

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**Q5** slide 9, note

You call spherical aberration the most challenging to deal with in high-resolution microscopy. In the IAL specifically, what causes it most often — coverslip thickness, wrong immersion medium, imaging depth, temperature, or something else? A rough ranking is enough.

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**Q6** slide 9 / objective markings

How much does using a #1 coverslip instead of a #1.5 actually cost at high NA, in your experience? Is it visible, or only measurable?

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## SECTION 2 – THE MISCONCEPTIONS YOU ALREADY NAME

**Q7** slides 28 and 32, notes

You write that people routinely close the aperture diaphragm or lower the condenser to gain contrast, and that this costs resolution and sharpness. How often do you see this in the IAL, and what tips you off from the image alone?

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**Q8** slides 28 and 32, notes

What do you say to the student who replies “but I can see it better this way”? We would like to use your answer rather than invent one.

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**Q9** slide 40, note

You are emphatic that DIC relief is optical rather than geometric. What is the worst real misinterpretation of that you have seen? A concrete case would be worth more to students than the principle.

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**Q10** slide 40, note

How do you demonstrate that the relief is optical? Rotating the specimen, rotating the prism, something else?

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**Q11** slide 22, note

Resolution follows NA and not magnification is the headline claim. Where do you find students first genuinely believe it — and is there a demonstration that does it faster than the argument?

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### SECTION 3 — PRACTITIONER'S KNOWLEDGE THAT IS IN NO TEXTBOOK

**Q12** slide 28, note

You note that Hoffman modulation contrast can see through birefringent plastic where DIC gives odd effects. Is Hoffman available on any IAL stand? If not, what do you tell live-cell people working in plastic dishes?

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**Q13** slide 36, note

You mention the phase ring is grey and can be seen through the back of the objective. Do you have students actually do this, and is there a stand where it is easiest?

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**Q14** slide 45, note

On full well capacity and binning — when do you tell someone to bin, and what is the tell that they should have been binning and were not?

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**Q15** slide 44, note

You point out upsampling smoothing pixels in the static image. Do you see people fooled by interpolated images into believing they have resolution they do not?

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**Q16** slide 50, note

What is your rule for when 8-bit is actually disqualifying rather than merely inadvisable?

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#### SECTION 4 — SAMPLING

**Q17** slide 45 body text; z-stack note

You recommend sampling at  $2.3\times$  the highest frequency rather than the textbook  $2\times$ , and  $2.3\times$  smaller than the axial resolution for z-steps. Where does 2.3 come from, and what did you see that made you settle on it?

*This is the kind of number students will otherwise round back down to 2, so the reason matters.*

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**Q18** z-stack note

You mention the Zeiss confocals and the M.2 have built-in calculators. Do students actually use them, and do they trust them appropriately?

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#### SECTION 5 — THE ABBE APPARATUS AND THE BACK FOCAL PLANE

**Q19** slide 23, note

Describing the Bertrand lens you write that rather than looking at the object, we observe what the object has done to the light traversing it. That is the clearest statement of the idea we have found. May we use it verbatim, attributed to you, as the epigraph for the back focal plane companion?

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**Q20** slide 23, note

Does the IAL have a diffraction apparatus or a suitable grating specimen? Could participants run Abbe's own experiment rather than only hear about it? That would be the strongest single demonstration in the day.

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**Q21** slide 21, note

You end the Abbe limit slide on the 1994 prediction and 1999 demonstration of breaking it. How much super-resolution do you want in scope for a one-day workshop — a mention, a section, or out entirely?

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## SECTION 6 – FOR THE ASSESSMENT WE ARE BUILDING

**Q22** cross-cutting

Of these five — resolution and NA, sampling, contrast-mode artifacts, illumination, dynamic range — which do IAL users most often get wrong in practice? A rough ranking lets us weight the item bank toward what actually goes wrong.

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**Q23** cross-cutting

What is the single worst outcome you have personally seen from one of these errors? A real consequence, even anonymised, does more than any argument we can make.

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**Q24** cross-cutting

Is there advice you give every new IAL user that appears nowhere in the deck at all?

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**Q25** logistics

Are you willing to have the room's aggregate result projected live during the workshop? It is pedagogically stronger and socially riskier than showing each participant their own result privately.

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## Errata for your confirmation

Four small things, harmless in a live talk where you are narrating, but worth fixing before they are frozen into a printed handout. Each is verified against the file you sent.

On the timeline slide the Janssen compound microscope is dated 1590, while the note for that same slide gives 1595. Both dates appear in the deck; which do you want?

The timeline gives “Christian Huygens”, while a later slide correctly has “Christiaan”.

The 1957 confocal entry reads “Marvin Minski”; the spelling is Minsky.

The bit-depth table has lost its exponents in conversion, so  $2^8$  renders as “28” and  $2^4$  as “24”. This may be an artifact of how we read the file rather than a fault in yours — worth a glance at the original.

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*Prepared for review. Nothing derived from these answers will be distributed to participants without your sign-off.*